

ISLI BASHA

Software Engineer — Full-Stack · TypeScript · Node.js · React

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PROFESSIONAL SUMMARY

Software engineer with 3+ years of production experience across full-stack TypeScript/Node.js, Python, and systems programming in Rust. Built and shipped: a production webhook service handling live WhatsApp Business traffic (Node.js + Express + PostgreSQL, 95%+ test coverage); an AI study companion with PDF ingestion and LLM-powered tutoring (FastAPI + Claude); a cyberphysical anomaly detection system for smart-city infrastructure (Flask + REST API + TDD, 68 tests); and a Cloudflare edge security service compiled to WASM. CS graduate (Polis University, 2026) — comfortable owning features end-to-end, working across ambiguous requirements, and shipping clean, tested code fast.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, Python, Rust, Go, Kotlin
Frontend	React, Next.js, Svelte, SvelteKit, Tailwind CSS, Three.js, HTML, CSS
Backend	Node.js, Express, NestJS, Hono, Fiber (Go), FastAPI, Axum (Rust), tRPC, GraphQL, REST API, WebSockets
Databases	PostgreSQL, SQLite, Turso, Redis
Cloud & Infra	Docker, CI/CD, GitHub Actions, Cloudflare Workers, Cloudflare Pages, Vercel, WebAssembly
Testing	Test-Driven Development (TDD), unit testing, integration testing, 95%+ coverage workflows
AI / LLMs	Claude AI, MCP SDK, LLM integration, RAG, PDF parsing, OCR, Zod schema validation
Other	Git, pnpm Workspaces, Agile, REST API design, HMAC-SHA256, JWT, system design

PROJECTS

ME-DT Framework — Multi-Domain Anomaly Detection System	09/2025 – 06/2026
Bachelor Thesis · Polis University, Tirana github.com/IsliBasha/me-dt-framework	
- Designed and built a cyberphysical threat detection backend across transport, energy, water, and communications infrastructure using Isolation Forest — custom correction logic reduced DOS-event false positives across 4 infrastructure domains. - Surfaced detections through a Flask REST API with clear contracts; mapped findings to MITRE ATT&CK; categories; visualised live data on a real-time JavaScript dashboard. 68 TDD tests.	
<i>Python · Flask · REST API · PostgreSQL · Isolation Forest · MITRE ATT&CK; · JavaScript · TDD</i>	
mira-study — AI Study Companion	2026
github.com/IsliBasha/mira-study	
- Built a full-stack AI desktop app: FastAPI backend exposes a clean REST API consumed by a PyQt6 desktop frontend — upload PDFs, auto-generate flashcards and multiple-choice quizzes, and get step-by-step AI tutor explanations via Claude; analytics dashboard tracks score trends and weak topics. - PDF text extraction via PyMuPDF + pdfplumber + Tesseract OCR; SQLite persistence via SQLAlchemy; Albanian/English i18n toggle; full test suite across backend and API layers.	
<i>Python · FastAPI · PyQt6 · Claude AI · SQLite · PyMuPDF · Tesseract OCR · REST API · LLM Integration</i>	
WhatsApp Business Bot — Inventory Webhook Service	01/2026 – Present
github.com/IsliBasha/Whatsapp-Chatbot	
- Built and deployed a production full-stack webhook service integrating WhatsApp Business with an Odoo inventory backend over XML-RPC — Node.js + Express backend with PostgreSQL persistence, requests authenticated via HMAC-SHA256, rate-limited per IP, and intent-parsed by Claude AI for fuzzy product search. - Maintained 95%+ test coverage across unit, integration, and end-to-end webhook message test suites; service handles live production traffic with full observability of request lifecycle.	
<i>TypeScript · Node.js · Express · Claude AI · PostgreSQL · Odoo XML-RPC · REST API · HMAC-SHA256</i>	
rust-scraper — Production Rust Web Scraper	01/2026 – Present

github.com/IsliBasha/rust-scraper

- Architected an 11-crate Cargo workspace: hybrid HTTP/headless rendering, CSS selector and XPath extraction engines, Tokio async worker pool, per-host token-bucket rate limiting — delivered as a single 9 MB self-contained binary with 33 tests across 7 crates.
- Ratatui terminal dashboard + Axum SSE web dashboard; SQLite WAL-mode persistence with crash-safe frontier recovery — demonstrates deep systems design and ownership across the full stack.

Rust · Tokio · Axum · SQLite · WebAssembly · Async · System Design · CI/CD

cf-worker-rust — Edge Security Service

01/2026 – Present

github.com/IsliBasha/cf-worker-rust

- Rust security service compiled to WASM, deployed to Cloudflare Workers — JWT validation (HS256/RS256) and sliding-window rate limiting enforced at the network edge with zero cold-start. Rate-limit state in Cloudflare KV; eliminates an entire auth round-trip per protected request.

Rust · WebAssembly · Cloudflare Workers · JWT · Cloudflare KV · Security · Edge Computing

Other Projects

- **FloraCare** — Android plant care app; Kotlin + Jetpack Compose, Perennial species API, care scheduling. (github.com/IsliBasha/FloraCare)
- **publer-mcp** — TypeScript MCP server, 13 tools, three-package pnpm monorepo, Zod-validated shared types, full mock mode. (github.com/IsliBasha/publer-mcp)

WORK EXPERIENCE

Freelance Software Engineer

01/2022 – 12/2024

Self-Employed · Tirana, Albania

- Delivered full-stack web projects for 5+ independent clients — landing pages, e-commerce builds, and portfolio sites — owning requirements, timelines, and client revisions end to end.
- Integrated third-party REST APIs and payment services; deployed to Vercel and Cloudflare Pages; adopted React and Node.js on projects requiring dynamic functionality.

TypeScript · React · Node.js · REST API · PostgreSQL · Git · Vercel · Client Management

Web Developer

01/2021 – 12/2021

BS Concept · Tirana, Albania

- Shipped front-end features for client web projects in a small team, consistently meeting deadlines.
- Resolved cross-browser rendering bugs via DevTools; contributed through pull requests, code reviews, and Agile sprint task tracking.

HTML · CSS · JavaScript · Git · Code Review · Agile

EDUCATION

Bachelor's in Computer Science

09/2022 – 06/2026

Polis University · Tirana, Albania

- Thesis: ME-DT Framework — cyberphysical threat detection for smart-city infrastructure using machine learning anomaly detection.

High School Diploma — Electronics

09/2018 – 06/2022

Harry T. Fultz · Tirana, Albania